

PYTHON REASONING QUICK CHEATSHEET

LAST-MINUTE PLACEMENT EXAM REVISION (1-2 MIN SCAN)



BASIC SYNTAX & DATA TYPES



- Variables $v = \square \square$
- print() `print(" ", *, " ")`
- basic arithmetic (+, -, *, /)
- Type casting `int, fl, flor, bool`
- Data types (int, float, str, bool)



STRING MANIPULATION



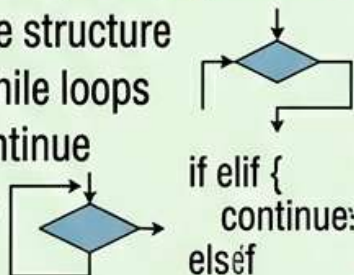
- Slicing `s[0:3]` $s = (a/s[0:3])$
- upper(), lower()
- split() $\text{split} \rightarrow$
- format() `code = "s[0:3]"`
- len() `len()`



CONTROL FLOW (If/Else, Loops)



- if, elif, else structure
- for and while loops
- break, continue
- Examples



LISTS, TUPLES, & DICTIONARIES



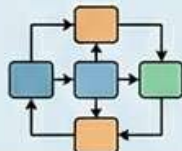
- List creation/slicing `[1] [1] ...`
- List methods: (append, pop, remove)
- Tuple syntax `A B C`
- Dictionary syntax and key-value access, popitem()



FUNCTIONS & MODULES



- def keyword
- def greet(name):
 return f"Hello {name}"
- parameters
- import, lamicd
- lambda functions



INPUT & EXCEPTION HANDLING

- input() for user data
- try/except blocks for error management
- def test_exception(x):
 try:
 y=1/x
 except: return 'Error'



ADVANCED CONCEPTS (OOP, Decorators)



- class syntax
 - methods
 - self
 - self
- Return ax methodd
- def test_decorator(func):
 wrap():
 func() wrap()
 wrap()
- list comprehensions
 [x for x in list]



BUILT-IN FUNCTIONS & LIBRARIES



- sum()
- max()
- min()
- range()
- sorted()
- import random,
 random.randint()
- import math,
 math.pi
- Revision zone recipe of counting (product rule)

